Analytic Number theory Homework

Problem 1. The Möbius function

(1) Let $n \geq 1$. For $n > 1$, write $n = p_1^{a_1} \cdots p_r^{a_r}$ $(a_j \geq 1)$. Express $\mu(n)$ in terms of a_j.

Prove that $\sum_{d|n} \mu(d) = \begin{cases} 1 & n=1 \\ 0 & n>1 \end{cases}$ hence, $|\mu(n)|$ is the indicator function of the squarefree functio

pf: [illegible]

Notice that $\mu(n) = \begin{cases} \frac{1}{2^r} \prod_{j=1}^{r} [(a_j - 2) - |a_j - 2|] & n > 1 \\ 1. & n = 1. \end{cases}$

$$\sum_{d|n} \mu(d) = 1 - \left(\sum_{p_i|n} 1 - \sum_{p_i p_j | n} 1 + \cdots + (-1)^r \sum_{p_1 p_2 \cdots p_r | n} 1 \right) = 1 - \left(C_r^1 - C_r^2 + \cdots + (-1)^{r-1} C_r^r \right) = 1 - C_r^0 = 0 \quad (n>1)$$

When $n = 1$. $\mu(n) = 1$.

(2) For real $s > 1$, use Euler products to prove $\sum_{n \geq 1} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}$.

Then prove $|\mu(n)| = \sum_{d^2|n} \mu(d)$ and deduce, for $x \geq 1$, $\sum_{n \leq x} |\mu(n)| = \frac{x}{\zeta(2)} + O(\sqrt{x})$.

pf: ① By definition, we immediately have $\sum_{n \geq 1} \frac{\mu(n)}{n^s}$ converges absolutely for real $s > 1$.

Then. $\sum_{n \geq 1} \frac{\mu(n)}{n^s} = \prod_p (1 - p^{-s})$ By Euler product: $\zeta(s) = \prod_p \left(\frac{1}{1 - p^{-s}} \right)$.

Thus. $\sum_{n \geq 1} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}$.

② $\sum_{d^2|n} \mu(d) = 1$ when $n = 1$.

$n > 1$ and be square-free, then $\sum_{d^2|n} \mu(d) = \mu(1) = 1$.

$n > 1$. and Not be square-free., then $n = p_1^{\alpha_1} \cdots p_s^{\alpha_s} p_{s+1} \cdots p_m$. $(\alpha_i \geq 2)$

$$\Rightarrow \sum_{d^2|n} \mu(d) = 1 - \left(\sum_{1 \leq i \leq s} 1 - \sum_{p_i^2 p_j^2 | n} 1 + \cdots + (-1)^{s-1} \sum_{p_1^2 \cdots p_s^2 | n} 1 \right) = 1 - C_s^0 = 0 = |\mu(n)|$$

③. $$\sum_{n \leq x} |\mu(n)| = \sum_{n \leq x} \sum_{d^2|n} \mu(d) = \sum_{1 \leq d \leq \lfloor \sqrt{x} \rfloor} \sum_{\substack{d^2|n \\ n \leq x}} \mu(d) = \sum_{1 \leq d \leq \lfloor \sqrt{x} \rfloor} \mu(d) \left\lfloor \frac{x}{d^2} \right\rfloor$$

$$= \sum_{1 \leq d \leq \lfloor \sqrt{x} \rfloor} \mu(d) \left(\frac{x}{d^2} + O(1) \right) \quad \left(\sum_{1 \leq d \leq \lfloor \sqrt{x} \rfloor} \mu(d) \, O(1) = O(\sqrt{x}) \right).$$

$$= x \sum_{d \geq 1} \frac{\mu(d)}{d^2} - x \sum_{d > \sqrt{x}} \frac{\mu(d)}{d^2} + O(\sqrt{x}) \overset{②}{=\!=} \frac{x}{\zeta(2)} + O(\sqrt{x})$$

(3). Let $M(x) = \sum_{n \leq x} \mu(n)$. Using summation by parts to prove, for real s,

$$\sum_{n \leq x} \frac{\mu(n)}{n^s} = M(x) \cdot x^{-s} + s \int_1^x \frac{M(t)}{t^{s+1}} \, dt.$$

Suppose that $M(x) = O(x^\theta)$ for real θ. Show that $\forall s > \theta$, $M(x) x^{-s} \to 0$ and

that $\int_1^\infty M(t)\,t^{-s-1}\,dt$ converges absolutely

pf: By summation by parts, we have $\sum_{n\le x}\frac{\mu(n)}{n^s} = \sum_{n\le x}\mu(n)\cdot\frac{1}{n^s} - \int_1^x M(t)\left(\frac{1}{t^s}\right)'dt$

$= M(x)\cdot x^{-s} + s\int_1^x \frac{M(t)}{t^{s+1}}\,dt$

Suppose that $M(x) = O(x^\theta)$ Then $\forall s>\theta$, $\int_1^\infty \frac{|M(t)|}{t^{s+1}}\,dt \le C\int_1^x t^{\theta-s-1}\,dt$.

let $x\to+\infty$ we have $\int_1^\infty \frac{|M(t)|}{t^{s+1}}\,dt$ converges $\square$

Rmk. $\forall\varepsilon>0$ $M(x) = O_\varepsilon(x^{\frac12+\varepsilon})$ $(x\to+\infty)$ is open, and equivalent to RH.

Problem 2. Euler's Totient Function.

(1). Prove, for $n\ge1$, $\varphi(n) = n\prod_{p\mid n}(1-\frac1p) = n\sum_{d\mid n}\frac{\mu(d)}{d}$.

pf: φ is a multiplicative function since there is an isomorphism $\mathbb{Z}_{mn}^* \cong \mathbb{Z}_m^*\times\mathbb{Z}_n^*$.

Thus $\varphi(n) = \varphi(p_1^{\alpha_1})\cdots\varphi(p_r^{\alpha_r}) = n\prod_{p\mid n}(1-\frac1p) \overset{\text{Simple Observation}}{=\!=} n\sum_{d\mid n}\frac{\mu(d)}{d}$.

(2) For $x\ge1$, prove $\sum_{n\le x}\varphi(n) = \frac{x^2}{2}\sum_{d\le x}\frac{\mu(d)}{d^2} + O(x\log(ex)) = \frac{x^2}{2\zeta(2)} + O(x\log(ex))$.

Deduce that, as $N\to\infty$ the proportion of ordered pairs $(a,b)\in[1,N]^2$ with $\gcd(a,b)=1$ tends to $\frac{1}{\zeta(2)}$.

pf: ① $\sum_{n\le x}\varphi(n) = \sum_{n\le x}n\sum_{d\mid n}\frac{\mu(d)}{d} = \sum_{1\le d\le x}\sum_{\substack{d\mid n\\ n\le x}}\frac{n\mu(d)}{d} = \sum_{1\le d\le x}\mu(d)\,\frac12\left\lfloor\frac{x}{d}\right\rfloor\left(\left\lfloor\frac{x}{d}\right\rfloor+1\right)$.

$= \frac{x^2}{2}\sum_{d\le x}\frac{\mu(d)}{d^2} - \sum_{d\le x}\frac{x}{2}\frac{\mu(d)}{d} + O(x)$ $\left(\left|\sum_{d\le x}\frac{\mu(d)}{d}\right| \le \sum_{d\le x}\frac1d = O(\log x)\right)$

$= \frac{x^2}{2}\sum_{d\le x}\frac{\mu(d)}{d^2} + O(\log x) + O(x) = \frac{x^2}{2}\cdot\sum_{d\le x}\frac{\mu(d)}{d^2} + O(x\log(ex)) = \frac{x^2}{2\zeta(2)} - \frac{x^2}{2}\sum_{d>x}\frac{\mu(d)}{d^2} + O(x\log(ex))$

$= \frac{x^2}{2\zeta(2)} + O(x\log(ex))$

② $\#\{(a,b)\in[1,N]^2 \mid \gcd(a,b)=1\} = N + \sum_{b\le N}\varphi(b) = N + \frac{N^2}{2\zeta(2)} + O(N\log(eN))$

(here $a\le b$, $a=1$)

Thus the proportion $= \frac{2}{N^2}\left(N + \frac{N^2}{2\zeta(2)} + O(N\log(eN))\right) \to \frac{1}{\zeta(2)}$

(3) Suppose a composite integer n satisfies $\varphi(n)\mid n-1$. Show that $2\nmid n$ and n is square-free

proof: $n = p_1^{\alpha_1}\cdots p_r^{\alpha_r} \Rightarrow p_1^{\alpha_1-1}\cdots p_r^{\alpha_r-1}(p_1-1)\cdots(p_r-1) \mid n-1$

If $2\mid n \Rightarrow 2\nmid n-1$ But $n = 2q$, $(q,2)=1$, $q>1 \Rightarrow \exists p\mid q$ $p\mid n-1 \Rightarrow 2\mid n-1$ Contradiction!

If $\exists p_i$ s.t. $p_i^2\mid n$, then $p_i\mid n-1$. Contradiction!

Rmk. Is there a composite integer n s.t. $\varphi(n)\mid n-1$? (Lehmer's totient problem)

Problem 3. Dirichlet L-function Let $q \geq 1$, Let χ be a Dirichlet character modulo q. and let χ_0 be principal modulo q

(1) For complex s with $\mathrm{Re}\, s > 1$, prove that the Dirichlet series converges absolutely and satisfies

$$L(s,\chi) = \sum_{n\geq 1} \frac{\chi(n)}{n^s} = \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}$$

For real $s > 1$, define $\log L(s,\chi) = \sum_p \sum_{k\geq 1} \frac{\chi(p^k)}{k p^{ks}}$. Show that the double series converges absolutely and that $\exp\left(\log(s,\chi)\right) = L(s,\chi)$.

For $1 < s < 2$, prove $\sum_p \frac{\chi(p)}{p^s} = \log(s,\chi) + O(1)$ where the implied constant is uniform, independent of s, q & χ

pf: $\sum_{n\geq 1} \left|\frac{\chi(n)}{n^s}\right| = \sum_{n\geq 1} \frac{1}{n^{\mathrm{Re}(s)}} < \infty$ ($\mathrm{Re}\, s > 1$) Thus $L(s,\chi)$ converges absolutely when $\mathrm{Re}\, s > 1$.

Then $\sum_{n\geq 1} \frac{\chi(n)}{n^s} = \sum_{n\geq 1} \frac{\chi(p_1^{\alpha_1}) \cdots \chi(p_s^{\alpha_s})}{p_1^{\alpha_1 s} \cdots p_s^{\alpha_s}} = \prod_p \left(1 + \left(\frac{\chi(p)}{p^s}\right)^1 + \left(\frac{\chi(p)}{p^s}\right)^2 + \cdots\right)$

$$= \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}$$

$\sum_p \sum_{k\geq 1} \left|\frac{\chi(p^k)}{k p^{ks}}\right| = \sum_p \sum_{k\geq 1} \frac{1}{k p^{ks}} < \sum_p \sum_{k\geq 1} \frac{1}{p^{ks}} = \sum_p \frac{1}{p^s - 1} < \infty$. Thus converges absolutely.

Use Euler product, $\log L(s,\chi) = -\sum_p \log\left(1 - \frac{\chi(p)}{p^s}\right) = \sum_p \sum_{k\geq 1} \frac{\chi(p^k)}{k p^{ks}} = L(s,\chi)$.

Thus. $\exp\left(\log(s,\chi)\right) = L(s,\chi)$.

$$\left|\sum_p \frac{\chi(p)}{p^s} - \sum_p \sum_{k\geq 1} \frac{\chi(p^k)}{k p^{ks}}\right| \leq \sum_p \sum_{k\geq 2} \left|\frac{\chi(p^k)}{k p^{ks}}\right| = \sum_p \sum_{k\geq 2} \frac{1}{k p^{ks}} < \sum_p \frac{1}{p^{2s-1}} < \infty \quad \square$$

(2). If $\chi \neq \chi_0$ and $A_\chi(t) = \sum_{n\leq t} \chi(n)$, prove $|A_\chi(t)| \leq q$ $(t \geq 1)$.

pf: Since $\chi \neq \chi_0$ $\sum_{0\leq n\leq q-1} \chi(n) = 0 \Rightarrow |A_\chi(t)| \leq q$. $\square$

Problem 4. Summation by parts.

Let $\{a_n\}_{n\geq 1} \subseteq \mathbb{C}$, $A(x) = \sum_{n\leq x} a_n$. $(x \geq 1)$.

(1) Let $x \geq 1$, and let $f : [1,x] \to \mathbb{C}$ be continuously differentiable. Prove that $\sum_{n\leq x} a_n f(n) = A(x)f(x) - \int_1^x A(t) f'(t)\, dt$

pf: $a_n = A(n) - A(n-1)$. Let $N = \lfloor x \rfloor$, then

then $\sum_{n=1}^{N} a_n f(n) = A(N)f(N) - \sum_{n=1}^{N-1} A(n)\left(f(n+1) - f(n)\right)$ (Abel)

$$= A(x)f(N) - \sum_{n=1}^{N-1} \int_n^{n+1} A(t) f'(t)\, dt = A(x)f(x) + A(x)\left(f(N) - f(x)\right) - \int_1^N A(t) f'(t)\, dt$$

$$= A(x)f(x) - \int_1^x A(t) f'(t)\, dt$$

(2). Suppose $|A(t)| \leq C t^\theta$ for $t \geq 1$. $C > 0$. $\theta \in \mathbb{R}$. For each $s > \theta$, take $f(t) = t^{-s}$. to prove that $\sum_{n\geq 1} a_n \cdot n^{-s}$ converges and that $\sum_{n>x} \frac{a_n}{n^s} = O_{C,s,\theta}\left(x^{\theta - s}\right)$.

if By summation by parts $\left|\sum_{n\le x}\frac{a_n}{n^s}\right| = \left|A(x)x^{-s} + s\int_1^x \frac{A(t)}{t^{s+1}}dx\right|$

$\le C x^{\theta-s} + s\int_1^x t^{\theta-s-1}dt = C x^{\theta-s} + \frac{s}{\theta-s}\left(x^{\theta-s}-1\right)$

$\Rightarrow \sum_{n\ge 1}\frac{a_n}{n^s}$ converges. $\sum_{n>x}\frac{a_n}{n^s} = O_{c,s,\theta}\left(x^{\theta-s}\right)$

(3) Let $a_n = \chi(n)$. ($\chi\neq\chi_0$) Show that $L(s,\chi)$ defined by (2) converges for each $s>0$. and satisfies $L(s,\chi) = s\int_1^\infty \frac{A_\chi(t)}{t^{s+1}}dt$. ($\chi\neq\chi_0$) Deduce, for $x\ge 1$, $L(1,\chi) = \sum_{n\le x}\frac{\chi(n)}{n} + O_q(x^{-1})$ and show that $L(s,\chi)\in C([0,+\infty))$

pf: ① $\forall s>0$ $\sum_{n\le x}\frac{\chi(n)}{n^s} = A_\chi(x)x^{-s} + s\int_1^x \frac{A_\chi(t)}{t^{s+1}}dt$ let $x\to+\infty$ by 3.(2). $|A_\chi(t)|\le q$.

we have $\sum_{n\ge 1}\frac{\chi(n)}{n^s}$ converges uniformly for $s>0$ Moreover, $\sum_{n\ge 1}\frac{\chi(n)}{n^s} = s\int_1^\infty \frac{A_\chi(t)}{t^{s+1}}dt$

② Then $L(1,\chi) = \int_1^\infty \frac{A_\chi(t)}{t^{s+1}}dt + \sum_{n>x}\frac{\chi(n)}{n^s}$

$= \sum_{n\le x}\frac{\chi(n)}{n} + O_q(x^{-1})$ By (2).

③ $L(s+\Delta s,\chi) - L(s,\chi) = (s+\Delta s)\int_1^\infty \frac{A_\chi(t)}{t^{s+\Delta s+1}}dt - s\int_1^\infty \frac{A_\chi(t)}{t^{s+1}}dt$

$= s\int_1^\infty \frac{A_\chi(t)}{t^{s+\Delta s+1}}\left(1-t^{\Delta s}\right)dt + \Delta s\int_1^\infty \frac{A_\chi(t)}{t^{s+\Delta s+1}}dt$

$= s\left(1-\xi^{\Delta s}\right)\int_1^\infty \frac{A_\chi(t)}{t^{s+\Delta s+1}}dt + \Delta s\int_1^\infty \frac{A_\chi(t)}{t^{s+\Delta s+1}}dt = P$ ($\Delta s\to 0$, we have $P\to 0$).

$\Rightarrow L(s,\chi)\in C([0,+\infty))$ □.

Problem 5. Nonvanishing of $L(1,\chi)$.

Fix $q\ge 3$. In part (2) – (4) assume that χ is real-valued and non-principal.

(1) For real $s>1$, use the identity

$$\sum_{\chi \bmod q}\operatorname{Log} L(s,\chi) = \varphi(q)\sum_{\substack{p \text{ prime},k\ge 1\\ p^k\equiv 1 \pmod q}}\frac{1}{k p^{ks}} \ge 0$$

Use that every nonprincipal $L(s,\chi)$ is analytic at $s=1$. Suppose that $\chi_1\neq\chi_2$. s.t $L(1,\chi_j)=0$

Show that, as $s\to 1^+$, that $L(s,\chi_j) = O_q(s-1)$, $L(s,\chi_0) = O_q\left((s-1)^{-1}\right)$.

Deduce that $\prod_{\chi \bmod q} L(s,\chi)\to 0$ $s\to 1^+$.

On the other hand, use 3.(1) we have $\prod_{\chi \bmod q} L(s,\chi)\ge 1$

Conclude that at most one nonprincipal character modulo q can satisfy $L(1,\chi)=0$ and that such character must be real-valued.

Pf: ① $\overline{\sum_{\chi \bmod q}} \log L(s,\chi) = \overline{\sum_{\chi \bmod q}} \sum_{p} \sum_{k\geq 1} \frac{\overline{\chi}(p^k)}{k\, p^{ks}}$

$= \sum_{p\nmid q} \sum_{k\geq 1} \overline{\sum_{\chi \bmod q}} \frac{\chi(p^k)}{k\, p^{ks}}$ (Next, we use $\overline{\sum_{\chi \bmod q}} \chi(n) = \begin{cases} \varphi(q) & n\equiv 1 \pmod q \\ 0 & n\not\equiv 1 \pmod q \end{cases}$

$= \varphi(q) \overline{\sum_{\substack{p^k\equiv 1 \pmod q \\ k\geq 1}}} \frac{\chi(p^k)}{k\, p^{ks}}$.

② $L(s,\chi)$